

2081

Bachelor of Science in Computer Science and Information Technology

Course Title: **Data Warehousing and Data Mining**

Code No: **CSC410** Semester: **7th Semester**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Section A

1. Define strong association rule. What are the limitations of APriori algorithm? Create a FP tree from the following data set. TIDList of Items T1{A, B, C} T2{B, C, D} T3{C, D} T4{B, D} T5{A, C} T6{A, C, D}
2. What is the role of Laplace smoothing? Create a decision tree from the following data set using ID3 as attribute selection approach. Object A1 A2 Class 1 TTC12 TTC13 TFC24 FFC15 FTC26 FTC2
3. Consider the data set (6,3), (7,2), (4,8), (2,2), (0,2), (9,0). Taking $k = 3$, show the result after first iteration using k -means algorithm. For choosing initial centroid, use k -means++ by taking (6, 3) as initial cluster center.

Section B

Attempt any Eight questions [8×5=40]

4. Explain about data mining primitives.
5. Define support vector. Write the algorithm for back propagation for classification.
6. What is data mart? Why do we need multidimensional data model?
7. Describe the different types of data object and attribute types.
8. What is data cube? List the different variations of cube materializations.
9. What is the concept behind beam search? Discuss about theory of balance and status.
10. Explain about web content, web usage and web structure mining.
11. Given the following distance matrix, find the core points and outliers using DBSCAN. Take $Eps = 2.5$ and $MinPts = 3$.
Data Points
A B C D E F G H A0 1.412.834.245.665.836.405.83 B0 1.412.824.244.475.004.47 C0 1.412.823.163.603.16 D0 1.412.002.242.00 E0 1.411.001.41 F0 1.002.82 G0 2.24 H0
12. List the components of data warehouse. Discuss about the trust propagation on social network.